2D Finite-Element Method Analysis of Rectangular, Circular, and Double-Ridged Waveguides

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***Abstract*—We developed a 2D finite-element eigenmode solver for empty, perfectly conducting, longitudinally uniform waveguides. For a homogeneous filling, the fields separate into TE and TM families, so the problem reduces to scalar eigenvalue equations for the longitudinal fields H_z and E_z. We use first-order triangular elements to produce sparse generalized eigenvalue systems whose eigenvalues give cutoff wavenumbers and whose eigenvectors give modal field distributions.**

We validate the solver on rectangular and circular waveguides by comparing the first three TE and TM modes with analytical results. We then re-work the solver to an empty double-ridged waveguide. The rectangular and circular results show close agreement with theory, while the ridged guide results show that the ridges lower the dominant TE cutoff and concentrate the fields in the narrow gap region.

***Index Terms*—Computational electromagnetics, finite element method, rectangular waveguide, circular waveguide, double-ridged waveguide.**

I. Introduction

WAVEGUIDE modal analysis is a classical problem in computational electromagnetics because the modal spectrum determines which field configurations can propagate, their cutoff frequencies, and how electromagnetic energy is distributed inside a guiding structure [2]. For hollow perfectly conducting (PEC) waveguides, canonical geometries such as the rectangular and circular guides are especially important because their modal solutions are known analytically. These geometries therefore provide natural benchmark problems for validating a numerical eigenvalue solver before that solver is applied to more complicated cross-sections [2].

Among numerical methods, the finite-element method (FEM) is particularly interesting for waveguide eigenvalue analysis because it handles irregular boundaries naturally while still producing sparse matrix systems suitable for efficient computation [1], [3]. In the special case considered here, namely an empty homogeneous waveguide with PEC walls, the vector Maxwell problem separates into TE and TM families. The unknowns can therefore be reduced to the scalar longitudinal fields H_z for TE modes and E_z for TM modes, which makes the formulation substantially simpler than the fully vectorial case [1]. This scalar reduction is one reason why empty PEC waveguides are a standard first test problem in FEM electromagnetics.

In addition to the canonical rectangular and circular guides, ridged waveguides are of practical interest because ridge loading reduces the dominant cutoff frequency and changes the field concentration in the guide interior, often increasing the useful single-mode bandwidth and strengthening the field in the gap region [4]–[7]. Unlike the rectangular and circular cases, however, the double-ridged waveguide does not generally admit a simple closed-form solution by separation of variables, so it is a natural example of a geometry for which a numerical eigenmode method is needed.

In this work, a 2D FEM solver based on first-order triangular elements is developed to compute the first three TE modes and first three TM modes of an empty rectangular waveguide and an empty circular waveguide. The rectangular guide is taken with aspect ratio $a/b = 2.2$, which is close to the requested 2:1 ratio but avoids exact degeneracy between TE_{20} and TE_{01}. The computed cutoff wavenumbers, mode shapes, and dispersion curves are then compared with analytical results. After this validation step, the same FEM formulation is applied to an empty double-ridged waveguide, and the corresponding modal fields and dispersion behavior are examined.

The remainder of this paper is organized as follows. Section II derives the TE/TM scalar eigenvalue problems from Maxwell's equations and presents the finite-element discretization. Section III summarizes the analytical formulas used to validate the numerical solver. Section IV presents the numerical results for the rectangular and circular guides, and Section V gives the corresponding results for the double-ridged guide. Finally, Section VI summarizes the main findings and discusses possible improvements and extensions.

II. Formulation and Discretization

We consider empty, perfectly conducting, longitudinally uniform waveguides whose cross-sections lie in the xy-plane and whose axis is the z-direction. The interior region Ω is filled by a single homogeneous, isotropic medium with constant permittivity ϵ and permeability μ, and the wall $\Gamma = \partial\Omega$ is modeled as a perfect electric conductor. The fields are assumed time harmonic with $e^{j\omega t}$ time dependence. Under these assumptions, the fields in a uniform guide can be expanded in modes with axial dependence $e^{-j\beta z}$, where β is the propagation constant. For an empty homogeneous single-conductor waveguide, the modes separate into transverse-electric and transverse-magnetic families, and no TEM mode exists [1], [2].

Inside the homogeneous source-free guide interior, Maxwell's equations are:

$$\nabla \times \mathbf{E} = -j\omega\mu\,\mathbf{H}, \tag{1}$$

$$\nabla \times \mathbf{H} = \quad j\omega\epsilon\,\mathbf{E}. \tag{2}$$

This is the authors second project for ECE 618.

The electric and magnetic fields are decomposed into transverse and longitudinal parts as:

$$\mathbf{E}(x,y,z) = [\mathbf{E}_t(x,y) + \hat{\mathbf{z}}E_z(x,y)]\, e^{-j\beta z}, \tag{3}$$
$$\mathbf{H}(x,y,z) = [\mathbf{H}_t(x,y) + \hat{\mathbf{z}}H_z(x,y)]\, e^{-j\beta z}. \tag{4}$$

It is convenient to split the gradient operator such that:

$$\nabla = \nabla_t + \hat{\mathbf{z}}\frac{\partial}{\partial z}, \qquad \nabla_t = \hat{\mathbf{x}}\frac{\partial}{\partial x} + \hat{\mathbf{y}}\frac{\partial}{\partial y}, \tag{5}$$

so that, under the modal dependence $e^{-j\beta z}$,

$$\frac{\partial}{\partial z} \mapsto -j\beta. \tag{6}$$

We also define the medium wavenumber:

$$k = \omega\sqrt{\mu\epsilon}, \tag{7}$$

and the cutoff wavenumber:

$$k_c^2 = k^2 - \beta^2. \tag{8}$$

therefore the modal dispersion relation is:

$$\beta^2 = k^2 - k_c^2. \tag{9}$$

To reduce the vector Maxwell system to scalar waveguide eigenproblems, we expand the curl operator under the modal ansatz. For any transverse vector field $\mathbf{F}_t(x,y)e^{-j\beta z}$,

$$\nabla \times \mathbf{F}_t = \nabla_t \times \mathbf{F}_t - j\beta\,\hat{\mathbf{z}} \times \mathbf{F}_t, \tag{10}$$

and for any longitudinal scalar field $\hat{\mathbf{z}}\phi(x,y)e^{-j\beta z}$,

$$\nabla \times (\hat{\mathbf{z}}\phi) = \nabla_t\phi \times \hat{\mathbf{z}} = -\hat{\mathbf{z}} \times \nabla_t\phi. \tag{11}$$

Therefore,

$$\nabla \times \mathbf{E} = \nabla_t \times \mathbf{E}_t - j\beta\,\hat{\mathbf{z}} \times \mathbf{E}_t - \hat{\mathbf{z}} \times \nabla_t E_z, \tag{12}$$
$$\nabla \times \mathbf{H} = \nabla_t \times \mathbf{H}_t - j\beta\,\hat{\mathbf{z}} \times \mathbf{H}_t - \hat{\mathbf{z}} \times \nabla_t H_z. \tag{13}$$

We begin with the TM family, for which

$$H_z = 0, \qquad E_z \neq 0. \tag{14}$$

Taking the transverse part of (2) and using (13) gives:

$$-j\beta\,\hat{\mathbf{z}} \times \mathbf{H}_t = j\omega\epsilon\,\mathbf{E}_t. \tag{15}$$

Similarly, taking the transverse part of (1) and using (12) gives:

$$-j\beta\,\hat{\mathbf{z}} \times \mathbf{E}_t - \hat{\mathbf{z}} \times \nabla_t E_z = -j\omega\mu\,\mathbf{H}_t. \tag{16}$$

Eliminating $\mathbf{H}_t$ from (15) and (16) yields the standard reconstruction formulas:

$$\mathbf{E}_t = -\frac{j\beta}{k_c^2}\nabla_t E_z, \tag{17}$$
$$\mathbf{H}_t = -\frac{j\omega\epsilon}{k_c^2}\hat{\mathbf{z}} \times \nabla_t E_z. \tag{18}$$

To obtain the scalar governing equation for E_z, we take the z-component of (2). Since $H_z = 0$, we have:

$$(\nabla \times \mathbf{H})_z = (\nabla_t \times \mathbf{H}_t)_z = j\omega\epsilon\, E_z. \tag{19}$$

Substituting (18) into (19) and using the transverse vector identity:

$$\nabla_t \times (\hat{\mathbf{z}} \times \nabla_t E_z) = \nabla_t^2 E_z$$

gives

$$\nabla_t^2 E_z + k_c^2 E_z = 0 \qquad \text{in } \Omega. \tag{20}$$

The PEC boundary condition for TM modes follows from the requirement that the tangential electric field vanish on the wall,

$$\hat{\mathbf{n}} \times \mathbf{E} = 0 \qquad \text{on } \Gamma. \tag{21}$$

Because the waveguide wall is parallel to the z-direction, the longitudinal electric field is tangential to the wall, so

$$E_z = 0 \qquad \text{on } \Gamma. \tag{22}$$

Thus the TM eigenproblem is:

$$\nabla_t^2 E_z + k_c^2 E_z = 0 \qquad \text{in } \Omega, \tag{23}$$
$$E_z = 0 \qquad \text{on } \Gamma. \tag{24}$$

For the TE family,

$$E_z = 0, \qquad H_z \neq 0. \tag{25}$$

Using the transverse part of (1) together with (12) we have:

$$-j\beta\,\hat{\mathbf{z}} \times \mathbf{E}_t = -j\omega\mu\,\mathbf{H}_t, \tag{26}$$

and the transverse part of (2) together with (13) gives

$$-j\beta\,\hat{\mathbf{z}} \times \mathbf{H}_t - \hat{\mathbf{z}} \times \nabla_t H_z = j\omega\epsilon\,\mathbf{E}_t. \tag{27}$$

Eliminating $\mathbf{E}_t$ yields:

$$\mathbf{H}_t = -\frac{j\beta}{k_c^2}\nabla_t H_z, \tag{28}$$
$$\mathbf{E}_t = \frac{j\omega\mu}{k_c^2}\hat{\mathbf{z}} \times \nabla_t H_z. \tag{29}$$

Taking the z-component of (1) then gives

$$(\nabla_t \times \mathbf{E}_t)_z = -j\omega\mu\, H_z, \tag{30}$$

and substituting (29) into (30) yields

$$\nabla_t^2 H_z + k_c^2 H_z = 0 \qquad \text{in } \Omega. \tag{31}$$

The TE boundary condition is again obtained from the PEC wall condition. Let $\hat{\mathbf{t}} = \hat{\mathbf{z}} \times \hat{\mathbf{n}}$ denote the unit tangent in the cross-section. Since $E_z = 0$ for TE modes,

$$\mathbf{E}_t \cdot \hat{\mathbf{t}} = 0 \qquad \text{on } \Gamma. \tag{32}$$

Using (29),

$$\begin{aligned}\mathbf{E}_t \cdot \hat{\mathbf{t}} &= \frac{j\omega\mu}{k_c^2}(\hat{\mathbf{z}} \times \nabla_t H_z) \cdot (\hat{\mathbf{z}} \times \hat{\mathbf{n}}) \\ &= \frac{j\omega\mu}{k_c^2}\,\nabla_t H_z \cdot \hat{\mathbf{n}} = \frac{j\omega\mu}{k_c^2}\,\partial_n H_z.\end{aligned} \tag{33}$$

Therefore,

$$\partial_n H_z = 0 \qquad \text{on } \Gamma. \tag{34}$$

The TE eigenproblem is:

$$\nabla_t^2 H_z + k_c^2 H_z = 0 \qquad \text{in } \Omega, \tag{35}$$
$$\partial_n H_z = 0 \qquad \text{on } \Gamma. \tag{36}$$

Because this is a pure Neumann eigenproblem, the constant function belongs to the kernel and produces the eigenvalue

$k_c = 0$. This null mode must be discarded numerically when extracting the physical TE spectrum [1].

The numerical solution is based on the weak forms of (23)–(24) and (35)–(36). For TM modes, we multiply (20) by a test function w and integrate over the cross-section:

$$\int_\Omega w \left(\nabla_t^2 E_z + k_c^2 E_z\right) d\Omega = 0. \tag{37}$$

Applying Green's identity we get:

$$-\int_\Omega \nabla_t w \cdot \nabla_t E_z \, d\Omega + \int_\Gamma w \, \partial_n E_z \, d\Gamma + k_c^2 \int_\Omega w E_z \, d\Omega = 0. \tag{38}$$

Since the TM boundary condition is essential, the admissible space is $H_0^1(\Omega)$ and the test function vanishes on Γ. The boundary term is therefore zero, and the weak form is: find nonzero $E_z \in H_0^1(\Omega)$ such that

$$\int_\Omega \nabla_t w \cdot \nabla_t E_z \, d\Omega = k_c^2 \int_\Omega w E_z \, d\Omega \qquad \forall w \in H_0^1(\Omega). \tag{39}$$

For TE modes, we multiply (31) by a test function w and integrate:

$$\int_\Omega w \left(\nabla_t^2 H_z + k_c^2 H_z\right) d\Omega = 0. \tag{40}$$

Again applying Green's identity we get:

$$-\int_\Omega \nabla_t w \cdot \nabla_t H_z \, d\Omega + \int_\Gamma w \, \partial_n H_z \, d\Gamma + k_c^2 \int_\Omega w H_z \, d\Omega = 0. \tag{41}$$

Since $\partial_n H_z = 0$ on Γ, the weak form becomes: find nonzero $H_z \in H^1(\Omega)$ such that

$$\int_\Omega \nabla_t w \cdot \nabla_t H_z \, d\Omega = k_c^2 \int_\Omega w H_z \, d\Omega \qquad \forall w \in H^1(\Omega). \tag{42}$$

The cross-section is discretized by a conforming triangular mesh as seen in Figs. 1, 4, and 7.

$$\Omega_h = \bigcup_{e=1}^{N_{\mathrm{el}}} \Omega^{(e)}. \tag{43}$$

For each triangle $\Omega^{(e)}$, let the local node coordinates be:

$$(x_1, y_1), \qquad (x_2, y_2), \qquad (x_3, y_3).$$

Using first-order nodal basis functions, the local approximation on each element is

$$u_h^{(e)}(x, y) = \sum_{i=1}^{3} u_i^{(e)} N_i^{(e)}(x, y), \tag{44}$$

where u_h denotes either E_z^h or H_z^h depending on the mode family. The standard P_1 triangular basis functions have the affine form:

$$N_i^{(e)}(x, y) = \frac{1}{2A^{(e)}} \left(a_i^{(e)} + b_i^{(e)} x + c_i^{(e)} y\right), \qquad i = 1, 2, 3, \tag{45}$$

where the element area is:

$$A^{(e)} = \frac{1}{2} \begin{vmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ 1 & x_3 & y_3 \end{vmatrix}, \tag{46}$$

and the coefficients are:

$$a_1^{(e)} = x_2 y_3 - x_3 y_2, \quad b_1^{(e)} = y_2 - y_3, \quad c_1^{(e)} = x_3 - x_2, \tag{47}$$

$$a_2^{(e)} = x_3 y_1 - x_1 y_3, \quad b_2^{(e)} = y_3 - y_1, \quad c_2^{(e)} = x_1 - x_3, \tag{48}$$

$$a_3^{(e)} = x_1 y_2 - x_2 y_1, \quad b_3^{(e)} = y_1 - y_2, \quad c_3^{(e)} = x_2 - x_1. \tag{49}$$

Since the basis functions are linear, their gradients are constant on each element:

$$\nabla_t N_i^{(e)} = \frac{1}{2A^{(e)}} \begin{bmatrix} b_i^{(e)} \\ c_i^{(e)} \end{bmatrix}. \tag{50}$$

The global finite-element approximation is assembled from these local basis functions. For TM modes, the expansion is written only over the free-node set N_E because the PEC boundary nodes are removed:

$$E_z^h(x, y) = \sum_{j \in N_E} u_j N_j(x, y). \tag{51}$$

For TE modes, all nodes remain unknown:

$$H_z^h(x, y) = \sum_{j \in N} u_j N_j(x, y). \tag{52}$$

We sub. these approximations into the weak forms and taking the Galerkin choice for the test functions, we get the generalized matrix eigenproblem:

$$[K]\mathbf{u} = k_c^2 [M]\mathbf{u}, \tag{53}$$

with stiffness and mass matrices:

$$K_{ij} = \int_\Omega \nabla_t N_i \cdot \nabla_t N_j \, d\Omega, \tag{54}$$

$$M_{ij} = \int_\Omega N_i N_j \, d\Omega. \tag{55}$$

These global matrices are assembled element by element. On a single triangle, the element stiffness matrix is:

$$[K^{(e)}]_{ij} = \int_{\Omega^{(e)}} \nabla_t N_i^{(e)} \cdot \nabla_t N_j^{(e)} \, d\Omega. \tag{56}$$

Because the gradients are constant on each element, (50) gives

$$[K^{(e)}]_{ij} = \frac{b_i^{(e)} b_j^{(e)} + c_i^{(e)} c_j^{(e)}}{4A^{(e)}}. \tag{57}$$

therefore:

$$[K^{(e)}] = \frac{1}{4A^{(e)}} \begin{bmatrix} b_1^2 + c_1^2 & b_1 b_2 + c_1 c_2 & b_1 b_3 + c_1 c_3 \\ b_2 b_1 + c_2 c_1 & b_2^2 + c_2^2 & b_2 b_3 + c_2 c_3 \\ b_3 b_1 + c_3 c_1 & b_3 b_2 + c_3 c_2 & b_3^2 + c_3^2 \end{bmatrix} \tag{58}$$

Similarly, the element mass matrix is:

$$[M^{(e)}]_{ij} = \int_{\Omega^{(e)}} N_i^{(e)} N_j^{(e)} \, d\Omega, \tag{59}$$

which for first-order triangles evaluates to:

$$[M^{(e)}] = \frac{A^{(e)}}{12} \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}. \tag{60}$$

Let element e connect global nodes $n_1^{(e)}, n_2^{(e)}, n_3^{(e)}$. The assembly process is:

$$K_{n_p^{(e)}, n_q^{(e)}} \mathrel{+}= [K^{(e)}]_{pq}, \tag{61}$$

$$M_{n_p^{(e)}, n_q^{(e)}} \mathrel{+}= [M^{(e)}]_{pq}, \tag{62}$$

for $p, q \in \{1, 2, 3\}$. This produces sparse global matrices because each nodal basis function is supported only on the small set of triangles attached to that node.

The boundary treatment differs between TM and TE. For TM modes, the Dirichlet condition $E_z = 0$ is imposed strongly by eliminating all nodes on the PEC boundary. If N_D is the set of Dirichlet nodes and N_E is the set of free nodes, then the system actually solved is:

$$[K]_{EE}\mathbf{u}_E = k_c^2 [M]_{EE}\mathbf{u}_E. \tag{63}$$

For TE modes, the Neumann condition is natural, so all nodes remain in the system and the full eigenproblem

$$[K]\mathbf{u} = k_c^2 [M]\mathbf{u} \tag{64}$$

is solved. The null eigenpair with $k_c = 0$ is then discarded.

Once the eigenvalues and eigenvectors are obtained, the cutoff quantities follow from

$$k_c = \sqrt{k_c^2}, \tag{65}$$

and the propagation constant is computed from:

$$\beta = \sqrt{k^2 - k_c^2}. \tag{66}$$

For vacuum filling, $k = k_0 = 2\pi f / c_0$, so

$$\beta(f) = \sqrt{\left(\frac{2\pi f}{c_0}\right)^2 - k_c^2}, \tag{67}$$

and the cutoff frequency is:

$$f_c = \frac{c_0 k_c}{2\pi}. \tag{68}$$

If $f < f_c$, then $\beta = j\alpha$ is purely imaginary and the mode is evanescent; if $f > f_c$, then β is real and the mode propagates. The dispersion curves in the results section are generated by evaluating (67) over a frequency grid after solving the geometry-dependent eigenproblem once.

The computed eigenvector gives the longitudinal field directly. The transverse fields are then reconstructed from the longitudinal field gradients. For TM modes,

$$\mathbf{E}_t = -\frac{j\beta}{k_c^2}\nabla_t E_z^h, \tag{69}$$

$$\mathbf{H}_t = -\frac{j\omega\epsilon}{k_c^2}\hat{\mathbf{z}} \times \nabla_t E_z^h, \tag{70}$$

and for TE modes,

$$\mathbf{H}_t = -\frac{j\beta}{k_c^2}\nabla_t H_z^h, \tag{71}$$

$$\mathbf{E}_t = \frac{j\omega\mu}{k_c^2}\hat{\mathbf{z}} \times \nabla_t H_z^h. \tag{72}$$

Because first-order elements are used, the longitudinal field is piecewise linear and continuous, while its gradient is piecewise constant on each element. As a result, the directly reconstructed transverse fields are piecewise constant element by element.

The same formulation is used for the rectangular, circular, and double-ridged guides. The difference lies only in the domain geometry and the associated mesh generation. For the rectangular guide, the computational domain is:

$$\Omega_{\text{rect}} = \{(x, y) : 0 < x < a,\ 0 < y < b\},$$

and the mesh is generated from a structured set of $(n_x \times n_y)$ points followed by Delaunay triangulation. For the circular guide, the computational domain is:

$$\Omega_{\text{circ}} = \{(x, y) : x^2 + y^2 < r^2\},$$

and the mesh is generated from concentric rings followed by Delaunay triangulation and removal of triangles whose centroids lie outside the circle. For the double-ridged guide, the outer rectangle is the same as the rectangular case, but two metallic ridges are removed from the interior. If the ridge width is w_r, the ridge depth is h_r, and

$$s = \frac{a - w_r}{2},$$

then the lower ridge occupies:

$$R_{\text{bot}} = [s, s + w_r] \times [0, h_r],$$

the upper ridge occupies:

$$R_{\text{top}} = [s, s + w_r] \times [b - h_r, b],$$

and the air region is:

$$\Omega_{\text{dr}} = (0, a) \times (0, b) \setminus (R_{\text{bot}} \cup R_{\text{top}}). \tag{73}$$

The ridge gap is:

$$g = b - 2h_r. \tag{74}$$

In the implementation, the ridged mesh is constructed by forcing point lines at the ridge edges in both x and y, triangulating the resulting point set, and deleting triangles whose centroids fall inside the ridge metal. This makes the mesh conform to the outer walls and ridge faces, which are all PEC boundaries.

The assumptions and approximations behind the numerical model can now be stated clearly. Analytically, the scalar TE/TM reduction relies on a homogeneous interior, PEC walls, and longitudinal uniformity. If the guide were partially filled or inhomogeneous, hybrid modes would generally arise and the scalar formulation used here would no longer be sufficient [1]. Numerically, the main approximations are that the exact geometry is represented by a triangular mesh and that the longitudinal field is approximated with piecewise-linear basis functions. These are standard approximations in FEM waveguide analysis and lead to sparse generalized eigenvalue problems that are well suited to the extraction of the lowest few modal cutoff values [1], [3].

TABLE I: Rectangular-waveguide geometry and mesh parameters.

Parameter	Value	Note
a	2.2	broad dimension
b	1.0	narrow dimension
n_x	61	mesh points in x
n_y	29	mesh points in y
a/b	2.2	near-2:1 ratio

TABLE II: Circular-waveguide geometry and mesh parameters.

Parameter	Value	Note
r	1.0	guide radius
n_r	26	radial layers
n_θ	140	outer-ring angular samples
Mesh type	concentric-ring	Delaunay after point generation

III. Analytical Validation Data

For the rectangular guide we have:

$$k_c^2 = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2. \tag{75}$$

The TM longitudinal fields are:

$$E_z^{(mn)}(x,y) = \sin\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right), \qquad m,n \geq 1, \tag{76}$$

while the TE longitudinal fields are:

$$H_z^{(mn)}(x,y) = \cos\left(\frac{m\pi x}{a}\right)\cos\left(\frac{n\pi y}{b}\right), \qquad (m,n) \neq (0,0). \tag{77}$$

For the circular guide of radius r, the TM cutoff values come from:

$$J_m(k_c r) = 0, \tag{78}$$

and the TE cutoff values come from:

$$J_m'(k_c r) = 0. \tag{79}$$

Hence:

$$k_c = \frac{\chi_{mn}}{r} \quad \text{for TM}, \qquad k_c = \frac{\chi'_{mn}}{r} \quad \text{for TE}, \tag{80}$$

where χ_{mn} is the nth positive zero of J_m and χ'_{mn} is the nth positive zero of J_m'.

IV. Numerical Results for the Rectangular and Circular Guides

The numerical parameters used in the code are summarized in Tables I–IV. The medium is vacuum in all cases, so $\epsilon = \epsilon_0$ and $\mu = \mu_0$. The rectangular guide uses $(a,b) = (2.2, 1.0)$ with a structured-point mesh of $n_x = 61$ and $n_y = 29$, while the circular guide uses radius $r = 1.0$ with $n_r = 26$ radial layers and $n_\theta = 140$ angular samples on the outer ring. For the double-ridged case, the outer rectangle remains $(2.2, 1.0)$ and the ridge geometry is $(w_r, h_r) = (0.9, 0.32)$, which gives a gap of $g = 0.36$.

TABLE III: Double-ridged-waveguide geometry and mesh parameters.

Parameter	Value	Note
a	2.2	outer width
b	1.0	outer height
w_r	0.9	ridge width
h_r	0.32	ridge depth
$g = b - 2h_r$	0.36	ridge gap
n_x	81	mesh points in x
n_y	49	mesh points in y

TABLE IV: Boundary treatment and mode-labeling procedure used by the solver.

Case	TM treatment	TE treatment	Labeling
Rectangle	Dirichlet elimination	Neumann, discard null	analytical matching
Circle	Dirichlet elimination	Neumann, discard null	Bessel-root matching
Double-ridged	Dirichlet elimination	Neumann, discard null	FEM-order only

A. Rectangular guide

The rectangular-waveguide mode shapes are shown in Fig. 2, and the corresponding dispersion curves are shown in Fig. 3. The first-row TE panels in Fig. 2 show the expected cosine-type longitudinal magnetic fields. The dominant TE_{10} mode varies mainly along the broad wall, while TE_{20} introduces one additional nodal transition in x. The TE_{01} mode varies primarily along the narrow wall. Because the code uses $a/b = 2.2$ rather than exactly 2, the TE_{20} and TE_{01} cutoffs are close but not identical, which avoids an exact degeneracy and makes the numerical identification cleaner.

The second-row TM panels in Fig. 2 show the expected Dirichlet behavior, with the longitudinal electric field vanishing on all four walls. The modal ordering TM_{11}, TM_{21}, and TM_{31} is consistent with the analytical spectrum. In Fig. 3, the FEM curves lie almost directly on top of the analytical curves, which confirms that the computed cutoff wavenumbers are accurate and that the mode identification is correct. Figure 1 shows the triangular P_1 mesh used for the rectangular guide. Because the geometry is itself rectangular, the mesh is nearly uniform across the cross-section and is appropriate for this validation case.

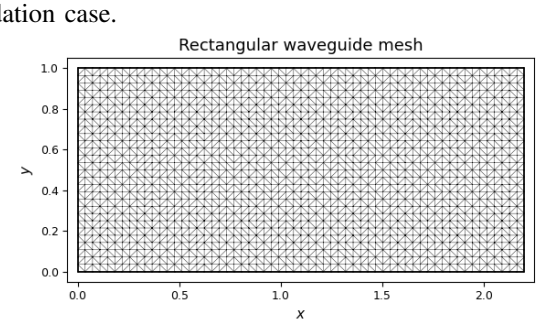

Fig. 1: Rectangular-waveguide triangular P_1 mesh used for the FEM computation.

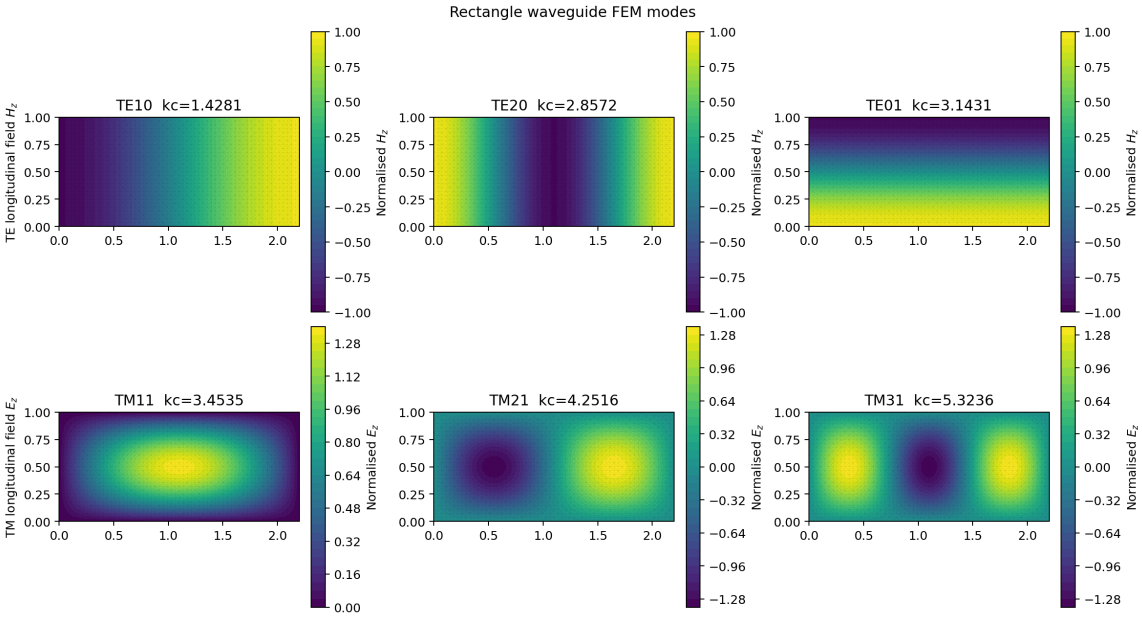

Fig. 2: Rectangular-waveguide FEM mode distributions. The top row shows the first three TE modes through the longitudinal magnetic field H_z, and the bottom row shows the first three TM modes through the longitudinal electric field E_z.

TABLE V: Rectangular-waveguide validation.

Mode	$k_{c,\mathrm{FEM}}$	$k_{c,\mathrm{exact}}$	$f_{c,\mathrm{FEM}}$ (GHz)	Rel. err.
TE_{10}	1.428144	1.427997	0.068142	0.0103%
TE_{20}	2.857169	2.855993	0.136325	0.0412%
TE_{01}	3.143069	3.141593	0.149967	0.0470%
TM_{11}	3.453549	3.450910	0.164781	0.0765%
TM_{21}	4.251611	4.245739	0.202859	0.1383%
TM_{31}	5.323627	5.312455	0.254009	0.2103%

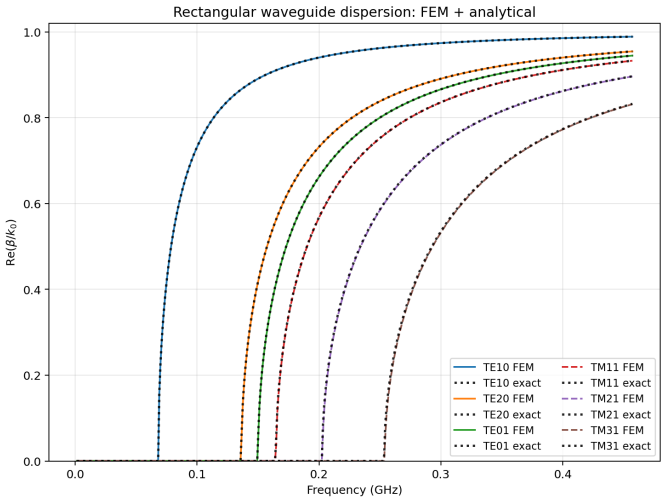

Fig. 3: Rectangular-waveguide dispersion curves. The FEM results closely follow the analytical curves over the full frequency range shown.

B. Circular guide

The circular-waveguide results are shown in Fig. 5 and Fig. 6. The plotted field patterns reproduce the expected Bessel-type structure. The dominant circular TE mode is

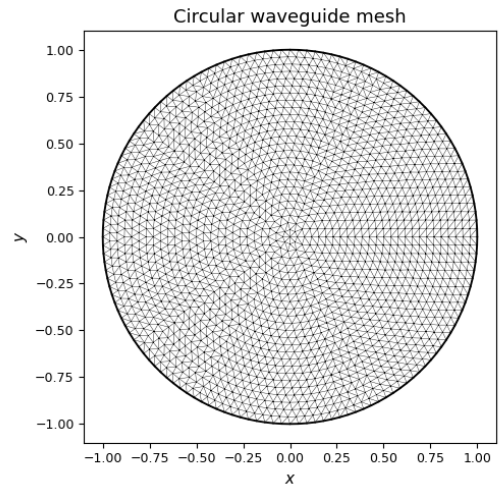

Fig. 4: Circular-waveguide triangular P_1 mesh used for the FEM computation.

TE_{11}, which varies primarily with azimuth and has one angular sign change across the diameter. The TE_{21} mode has higher angular order, while the azimuthally symmetric TE_{01} mode has stronger radial structure and no angular sign change. The TM fields follow the corresponding Dirichlet spectrum and again match the expected modal ordering.

The dispersion plot in Fig. 6 shows the same trend as in the rectangular case: the FEM curves are nearly indistinguishable from the analytical curves over the plotted range. This confirms that the circular-mesh generation, Bessel-root-based mode matching, and eigenvalue extraction are all behaving correctly.

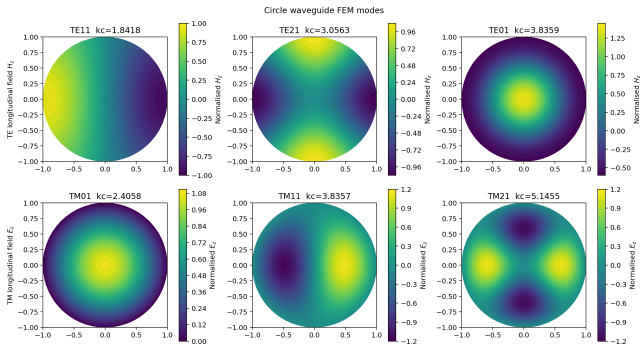

Fig. 5: Circular-waveguide FEM mode distributions. The top row shows the first three TE modes through the longitudinal magnetic field H_z, and the bottom row shows the first three TM modes through the longitudinal electric field E_z.

TABLE VI: Circular-waveguide validation.

Mode	$k_{c,\mathrm{FEM}}$	$k_{c,\mathrm{exact}}$	$f_{c,\mathrm{FEM}}$ (GHz)	Rel. err.
TE_{11}	1.841788	1.841184	0.087878	0.0328%
TE_{21}	3.056347	3.054237	0.145829	0.0691%
TE_{01}	3.835883	3.831706	0.183023	0.1090%
TM_{01}	2.405750	2.404826	0.114787	0.0384%
TM_{11}	3.835706	3.831706	0.183015	0.1044%
TM_{21}	5.145477	5.135622	0.245508	0.1919%

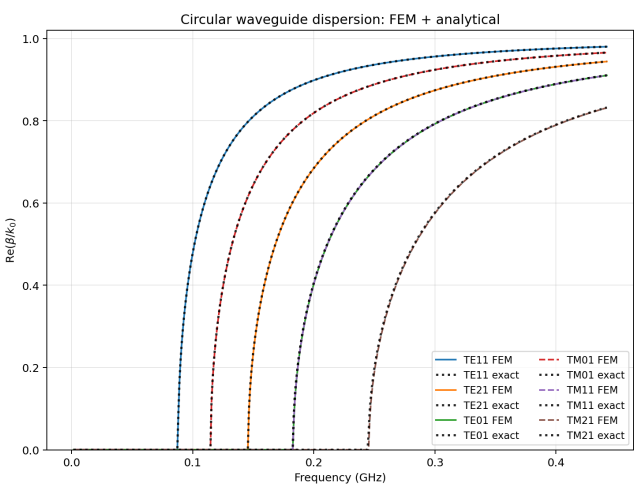

Fig. 6: Circular-waveguide dispersion curves. The FEM results closely match the analytical Bessel-root-based curves.

V. Results for the Double-Ridged Waveguide

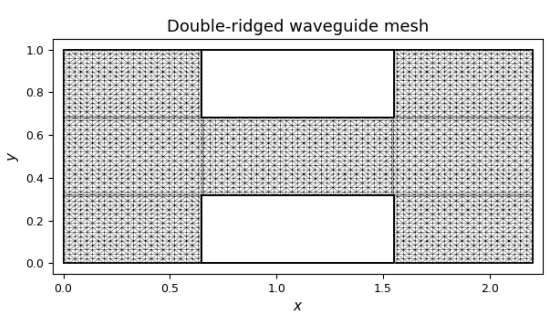

Fig. 7: Double-Ridged-waveguide triangular P_1 mesh used for the FEM computation.

TABLE VII: Double-ridged-waveguide cutoff data from the FEM solver.

Mode	k_c	f_c (GHz)	Mult.
TE mode 1	0.933568	0.044544	1
TE mode 2	2.793677	0.133296	1
TE mode 3	3.233124	0.154264	2
TM mode 1	5.483877	0.261655	2
TM mode 2	7.914735	0.377639	2
TM mode 3	8.740285	0.417029	1

The double-ridged guide uses the same outer dimensions as the rectangular case, but two metal ridges are inserted symmetrically from the top and bottom walls. In the code, the default geometry is

$$a = 2.2, \qquad b = 1.0, \qquad w_r = 0.9, \qquad h_r = 0.32,$$

which gives the ridge gap

$$g = b - 2h_r = 0.36.$$

The air-filled computational domain is therefore the outer rectangle with the two ridge rectangles removed.

The important point is that the governing TE and TM scalar eigenvalue problems do not change. The medium is still homogeneous, the guide is still uniform in z, and the wall is still PEC. The difference is entirely geometric. Because the gap is narrow and the ridge tips create abrupt geometric transitions, the ridged case benefits from a denser mesh than the plain rectangular guide.

The mode plots in Fig. 8 show this change in field behavior. The lowest TE mode has strong field concentration in the ridge-gap region, and the next two TE modes also place a large fraction of their energy in the central part of the guide. The TM modes occur at much larger cutoff wavenumbers than the dominant TE mode, which is consistent with the fact that the ridged geometry primarily lowers the dominant TE-type cutoff. This trend is also visible in Fig. 9, where the first TE branch begins propagating at a much lower frequency than the first TM branch.

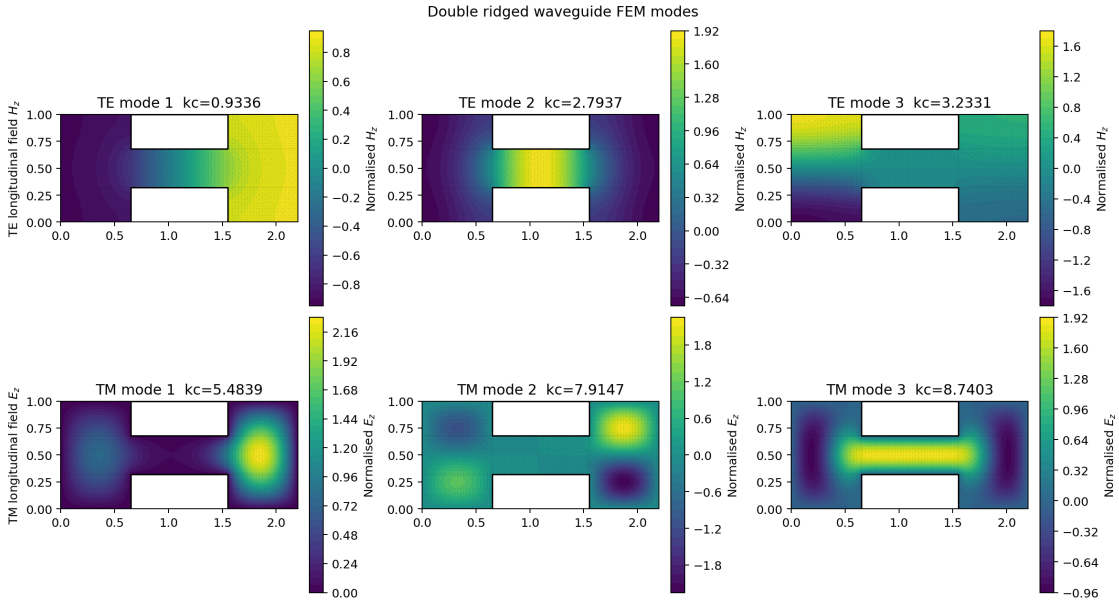

Fig. 8: Double-ridged-waveguide FEM mode distributions. The top row shows the first three TE modes through H_z, and the bottom row shows the first three TM modes through E_z. Ridge outlines are included to show the field concentration in the central gap region.

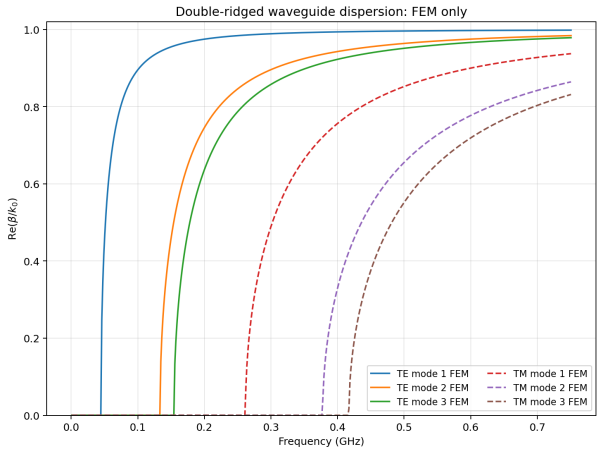

Fig. 9: Double-ridged-waveguide FEM dispersion curves. The dominant TE mode starts propagating at much lower frequency than the first TM mode, which is the expected effect of ridge loading.

VI. Conclusion

A 2D finite-element eigenmode solver was developed for empty homogeneous PEC waveguides and applied to rectangular, circular, and double-ridged cross-sections. Starting from Maxwell's equations, the problem was reduced to scalar TE and TM eigenvalue problems for the longitudinal fields H_z and E_z. These scalar problems were discretized with first-order triangular elements, which produced sparse generalized eigenvalue systems for the cutoff wavenumbers and corresponding eigenvectors for the mode shapes.

For the rectangular and circular guides, the numerical cutoff values and dispersion curves were found to agree very closely with analytical results. The field plots also showed the expected longitudinal modal structure, which supports the correctness of both the formulation and the implementation. After that validation step, the same finite-element framework was applied to the empty double-ridged guide. In that case the main difficulty is geometric rather than theoretical, since the governing equations and boundary conditions are unchanged. The results show that the ridges strongly reduce the dominant TE cutoff and concentrate the field in the narrow central gap.

Several natural extensions could improve the present solver. Higher-order triangular basis functions could reduce numerical error for a fixed mesh size. A fully vectorial formulation would also allow analysis of partially filled or inhomogeneous guides, where the scalar TE/TM reduction used here no longer applies. A nice extension would also include a GUI.

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